

Decision tree: Choosing a polygon operation

st_make_valid()

st_is_valid(reason = TRUE) names the fault; run the repair as part of reading a polygon file

then

What does the question need?

which features are involved

st_filter()

whole features, with their fields untouched

how much of a layer is inside another

st_intersection()

the geometry cut to the overlap; an area field has to be recalculated with st_area()

what each field means for its geometry

st_agr()

records constant, aggregate or identity; does not change the field's values

how much lies near the polygon boundary

st_cast() + st_buffer()

cast the boundary to a line, buffer it, then clip the result to the polygon

how to reduce the file size for drawing

st_simplify()

dTolerance in the units of the CRS; use ms_simplify() to keep shared boundaries matched